

User Acceptance Testing (UAT) Template: ShopEZ Stock Trading Platform

This document presents the testing parameters, test environments, detailed test scenarios, and bug tracking logs for the ShopEZ Stock Trading Platform.

Project Overview

- **Project Name:** ShopEZ Stock Trading Platform
 - **Project Description:** A full-stack MERN (MongoDB, Express, React, Node) application allowing users to view market tickers, observe simulated real-time price fluctuations, buy and sell mock equities with a virtual balance, track portfolio cost bases, and access administrative audits.
 - **Project Version:** 1.0.0
 - **Testing Period:** June 10, 2026 to June 17, 2026
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Testing Environment

- **Frontend Location:** Local Development URL `http://localhost:5173`
 - **Backend API Location:** Local API URL `http://localhost:5000/api`
 - **Pre-seeded Accounts (Credentials):**
 - **Standard Trader (USER Role):** `user@shopez.com` / `password123`
 - **Platform Administrator (ADMIN Role):** `admin@shopez.com` / `password123`
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Test Cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	User Registration (Positive)	1. Navigate to `/register`. 2. Fill in Name, unique email, password, and role "USER". 3. Click Register.	Account is created in MongoDB, password is encrypted with bcrypt, starting balance is set to \$10,000, empty portfolio is created, and user is redirected to dashboard.	Account successfully created, empty portfolio is instantiated, and redirected to `/`.	Pass
TC-002	User Registration (Negative)	1. Navigate to `/register`. 2. Enter an email that is already registered. 3. Click Register.	Registration fails. Server returns a 400 error status, and React client displays the validation message "User already exists".	Validation error message displayed, registration blocked.	Pass
TC-003	User Authentication Login	1. Navigate to `/login`. 2. Enter email `user@shopez.com` and password `password123`. 3. Click Sign In.	JWT token is returned from backend, stored in local storage, Axios headers default to `Bearer `, and user is authenticated.	Token saved, AuthContext state updated, and user redirected to Dashboard.	Pass
TC-004	Live Price Fluctuation	1. Open the Dashboard or Market page. 2. Observe listed stock prices for 10-15 seconds.	Stock prices change by -1.2% to +1.2% every 5 seconds, updating open, high, low, absolute change, and percentage stats in Mongoose.	Prices and change stats fluctuated dynamically in grid and charts.	Pass

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-005	Buy Stock Order (Positive)	1. Navigate to `/stocks/AAPL`. 2. Select "BUY" tab. 3. Enter quantity `10` and click BUY.	Virtual cash balance is deducted by cost (price * 10), AAPL holding is created/updated in portfolio with average cost basis, and a Transaction document is saved.	Cash balance deducted, holdings updated on UI in real-time, order successful.	Pass
TC-006	Buy Stock Order (Negative)	1. Navigate to `/stocks/NVDA`. 2. Enter quantity `50` (estimated cost exceeds current balance). 3. Click BUY.	Order fails. Server rejects transaction and returns a 400 error status with message "Insufficient balance", which is shown in red on the UI.	Error message shown, transaction blocked.	Pass
TC-007	Sell Stock Order (Positive)	1. Navigate to `/stocks/AAPL`. 2. Select "SELL" tab. 3. Enter quantity `5` (user owns 10 shares). 4. Click SELL.	Revenue is added to virtual balance, AAPL holding quantity decreases to 5, and a SELL Transaction record is logged.	Cash balance credited, owned shares updated, order successful.	Pass
TC-008	Sell Stock Order (Negative)	1. Navigate to `/stocks/AAPL`. 2. Select "SELL" tab. 3. Enter quantity `12` (user only owns 5 shares). 4. Click SELL.	Transaction is rejected. Server returns a 400 error status with message "Insufficient shares", and the transaction is blocked.	Error alert displayed, transaction rejected.	Pass

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-009	Admin Stock Listing Management	1. Log in as `admin@shopez.com`. 2. Go to Admin Panel → Manage Stocks. 3. Click List New Stock, enter `TSLA`, company name, price, and click Publish.	New stock ticker is added to MongoDB and starts fluctuating; immediately visible on Trader dashboard.	Stock successfully listed and visible across the app.	Pass
TC-010	Admin Global Audit Trails	1. Log in as `admin@shopez.com`. 2. Go to Admin Panel → Users Database. 3. Go to Admin Panel → Audit Logs.	Admin can view the names, emails, roles, balances, and asset valuations of all users, and trace all BUY/SELL trade actions.	Database listing and global transaction logs successfully rendered.	Pass

Bug Tracking Log

Bug ID	Bug Description	Steps to Reproduce	Severity	Status	Additional Feedback / Resolution
BG-001	Missing buy/sell input validation (zero/negative numbers).	1. Go to stock detail page. 2. Input <code>-5</code> in quantity. 3. Click BUY.	High	Closed	Added client-side html <code>min="0.0001"</code> attribute and server-side validation to reject quantities <code><= 0</code> .
BG-002	Route protection failure when accessing portfolio directly.	1. Log out. 2. Directly type <code>http://localhost:5173/portfolio</code> in browser.	High	Closed	Implemented <code>ProtectedRoute.jsx</code> client router guard to redirect unauthenticated requests to <code>/login</code> .
BG-003	Navbar dropdown rendering in dark colors.	1. Log in. 2. Click profile dropdown.	Low	Closed	Removed <code>dropdown-menu-dark</code> class from profile dropdown and added custom light-theme borders.

Sign-off

- **Tester Name:** Srinithya
- **Date:** 17 June 2026
- **Notes:** All core trading flows, real-time updates, role guards, and light-theme visual configurations have passed testing successfully.